03 - Image Modelling Baseline

This notebook trains and evaluates a minimal image-only baseline on the clean shared split. It compares a majority-class baseline with a pretrained ResNet18 whose classification head is trained for the 27 Rakuten product classes.

1. Evaluation Contract

Training uses only rows marked `train` in `image_split_clean.csv`. Scores are computed only on rows marked `validation`. The official challenge test set is excluded because it has no labels. Weighted F1 is the primary score and macro F1 is reported to expose performance on smaller classes.

2. Imports and Configuration

```
Device: mps
```

3. Load the Clean Manifest and Encode the Labels

The Rakuten class codes are mapped deterministically to indices from 0 to 26 for `CrossEntropyLoss`. Image paths are reconstructed locally and are not stored in the portable manifest.

	rows	classes
train	62312	27
validation	16984	27

4. Majority-Class Baseline

This baseline predicts the most frequent training class for every validation product. It provides the minimum reference score that the image model must improve upon.

```
Majority class: 2583
```

	accuracy	weighted_f1	macro_f1
majority_class	0.1202	0.0258	0.0080

5. Image Dataset and DataLoaders

The default ResNet18 preprocessing resizes, crops and normalizes images using the ImageNet configuration associated with the pretrained weights. The same deterministic transform is used for this first baseline.

```
Training batches: 1948
Validation batches: 531
```

6. Configure the ResNet18 Baseline

The pretrained feature extractor is frozen and only the final classification layer is trained. This keeps the first modelling iteration fast and provides a clear baseline before fine-tuning or augmentation.

```
Trainable parameters: 13851
```

7. Train the Classification Head

epoch	train_loss
1	1.8135

8. Evaluate and Save the Baseline

The checkpoint, label mapping and validation predictions are saved for reproducibility and later multimodal fusion. The prediction file retains `row_id` and all 27 class probabilities.

	accuracy	weighted_f1	macro_f1
majority_class	0.1202	0.0258	0.0080
resnet18_frozen	0.5016	0.4817	0.4230

```
Validation loss: 1.6695
```

	prdtypecode	precision	recall	f1	support
2	50	0.692	0.054	0.099	336
8	1281	0.312	0.060	0.101	414
24	2585	0.292	0.066	0.108	499
17	2220	0.579	0.067	0.120	165
6	1180	0.301	0.144	0.195	153
11	1302	0.316	0.157	0.209	498
7	1280	0.315	0.352	0.333	974
12	1320	0.293	0.471	0.361	648
0	10	0.519	0.281	0.365	623
22	2582	0.324	0.425	0.367	518
1	40	0.354	0.384	0.369	502
16	2060	0.526	0.308	0.389	999
10	1301	0.580	0.317	0.410	161
20	2462	0.397	0.447	0.421	284
26	2905	0.545	0.385	0.451	174
13	1560	0.468	0.514	0.490	1015
4	1140	0.465	0.573	0.513	534
21	2522	0.484	0.570	0.524	998
19	2403	0.591	0.479	0.529	955
15	1940	0.652	0.453	0.535	161
3	60	0.471	0.675	0.554	166
9	1300	0.473	0.731	0.574	1009
18	2280	0.475	0.804	0.597	952
25	2705	0.577	0.634	0.604	552
23	2583	0.668	0.666	0.667	2042
14	1920	0.683	0.760	0.719	861
5	1160	0.827	0.808	0.817	791